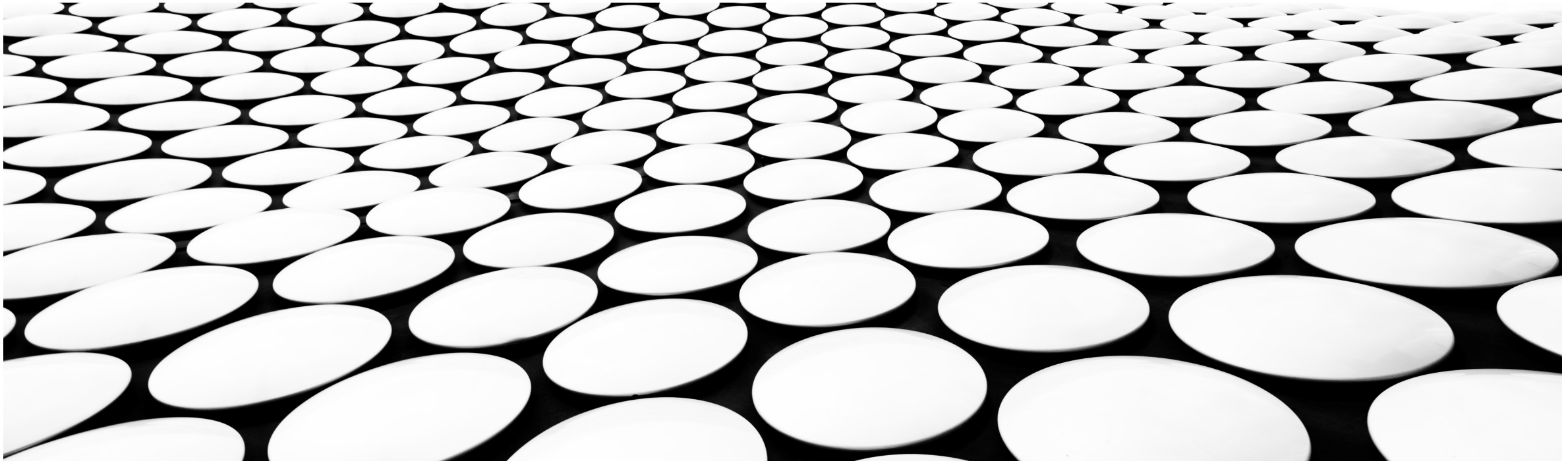
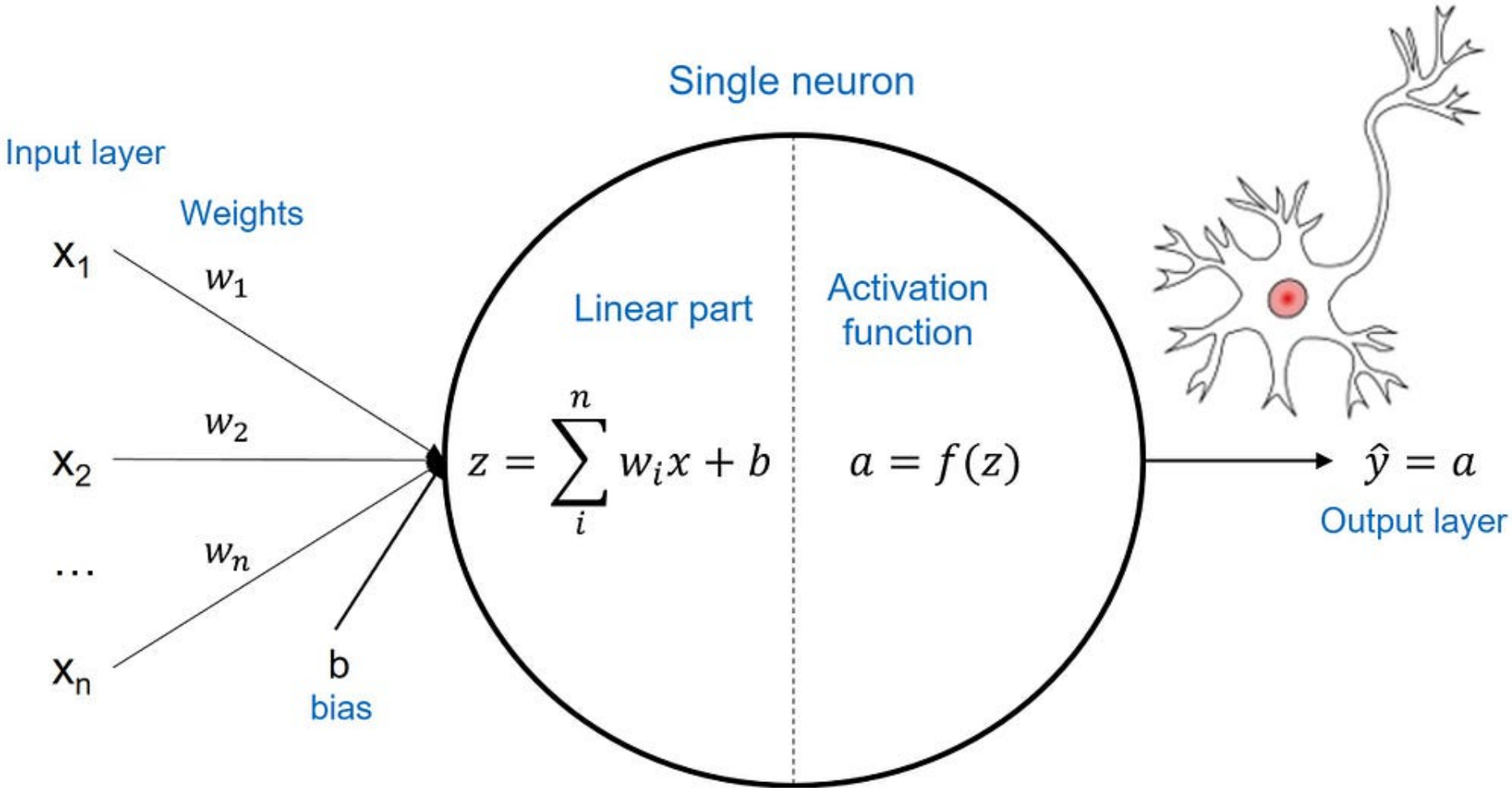
DEEP NEURAL NETWORKS

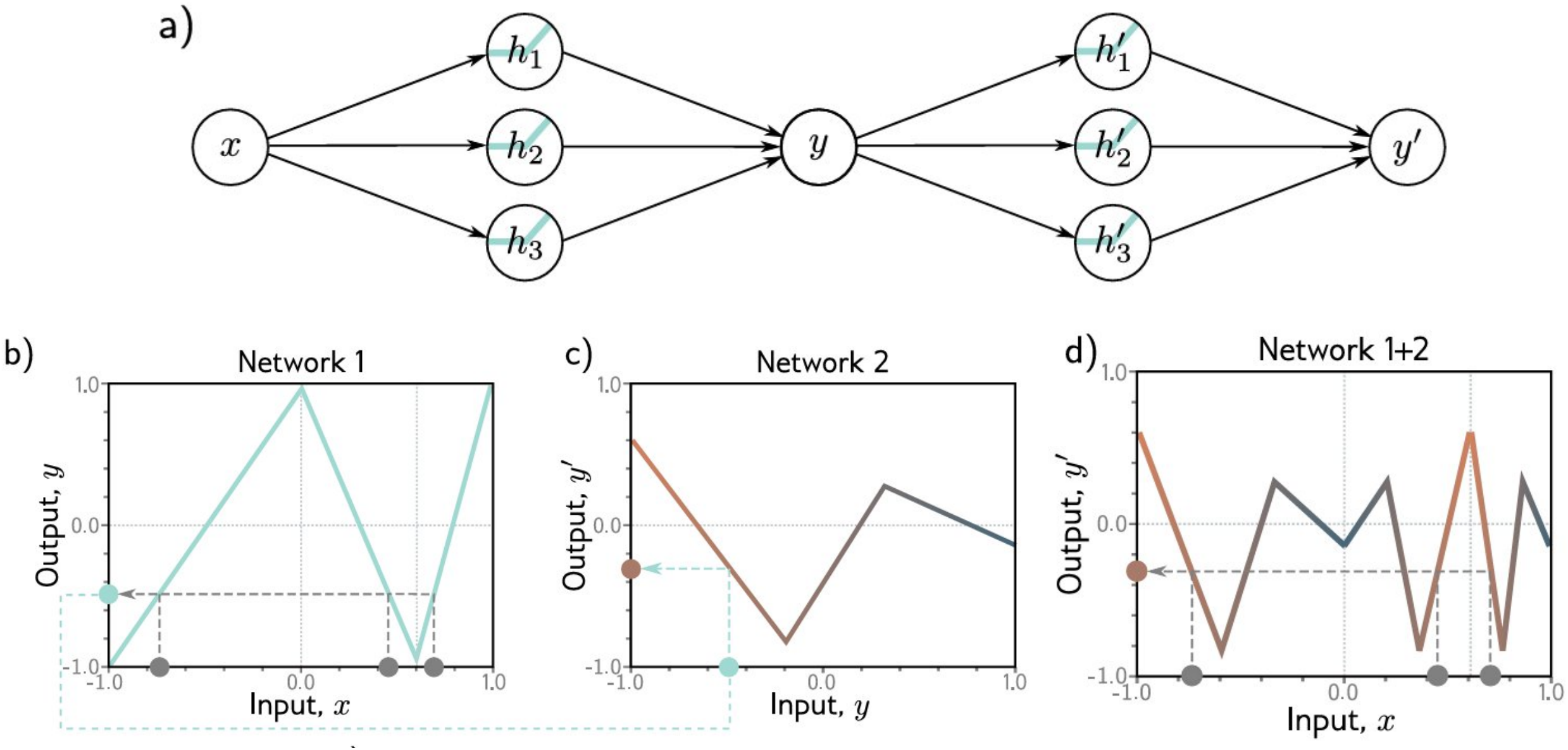
FROM NEURONS TO GRADIENT DECENT

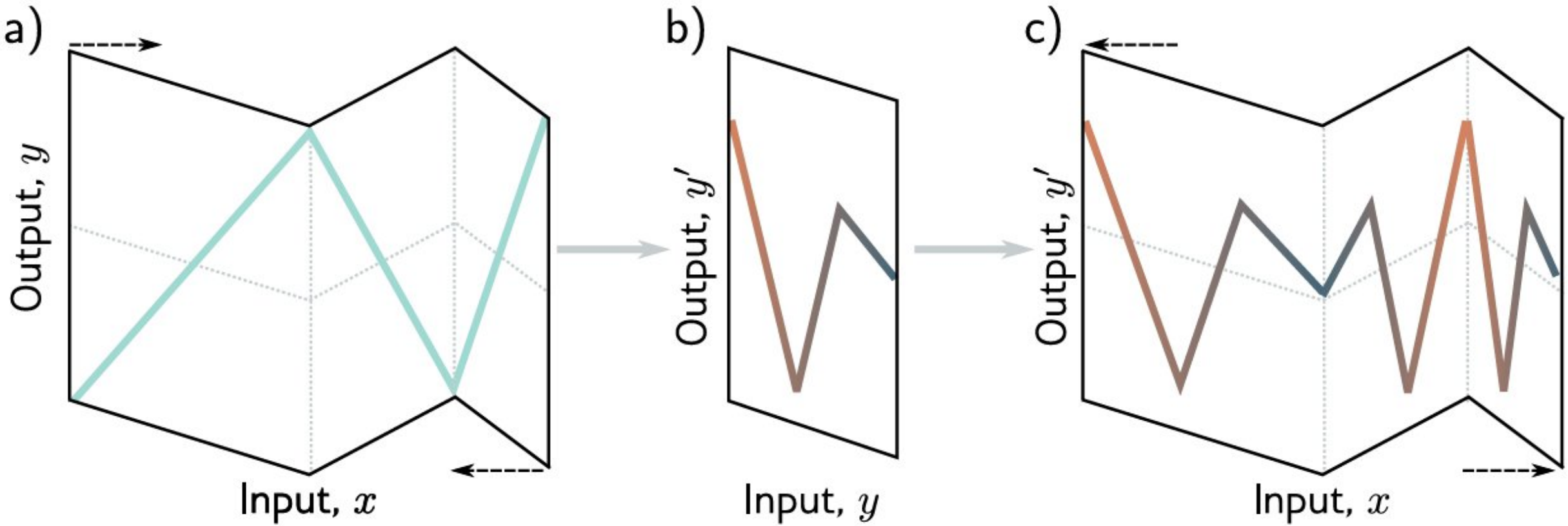


CHAPTER 4:
DEEP NEURAL NETWORKS

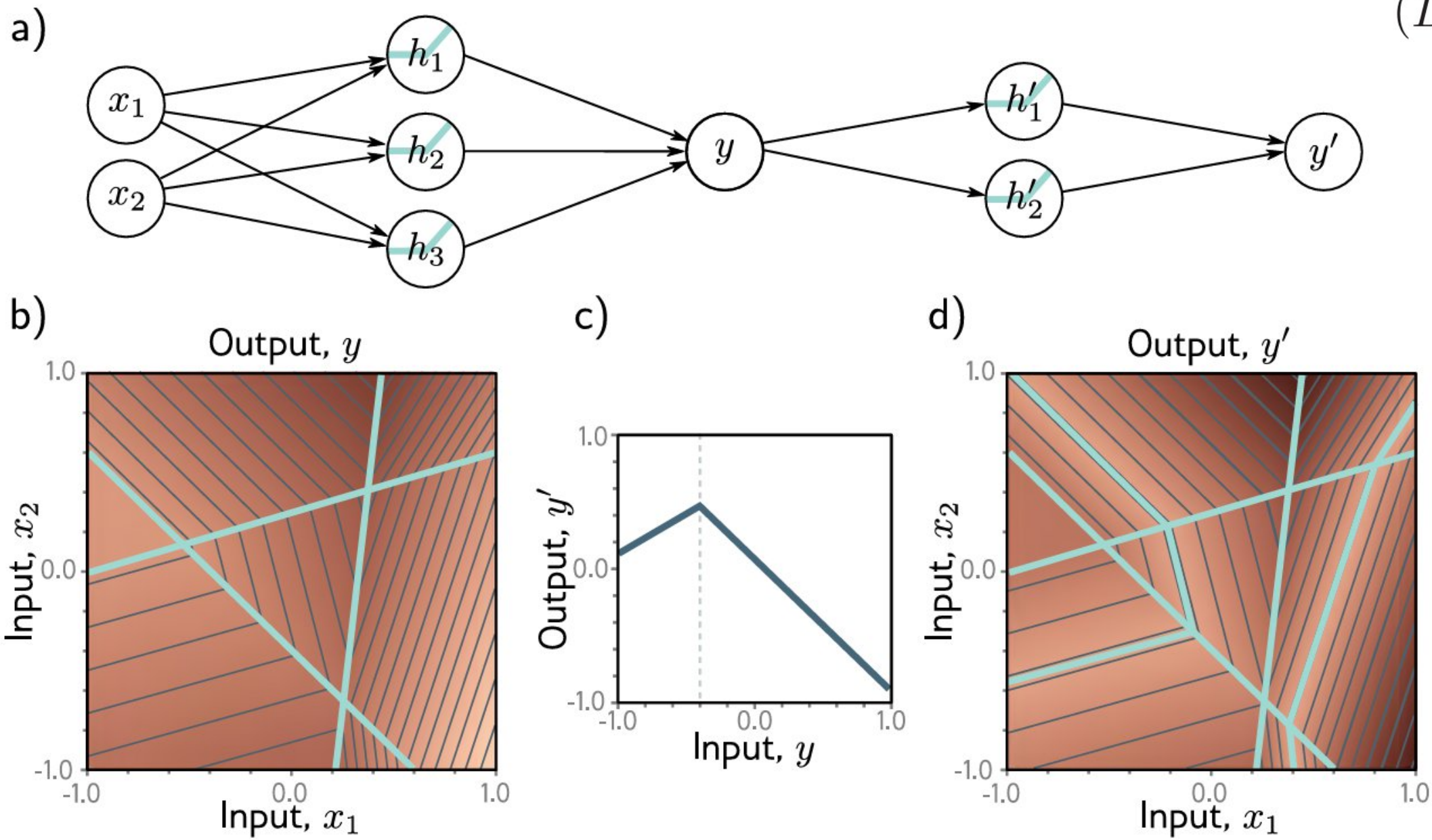
A SINGLE NEURON

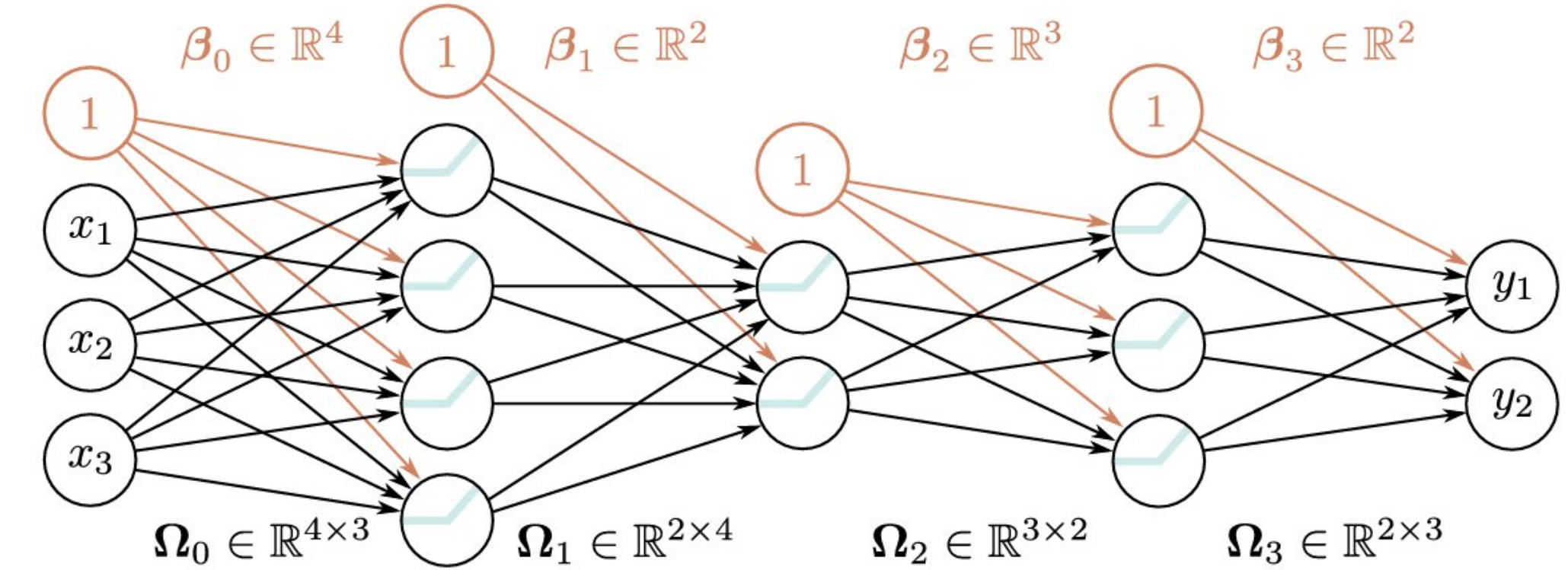



a)
x
h_1
h_2
h_3
y
h'_1
h'_2
h'_3
y'
b)
Network 1
Output, y
Input, x
c)
Network 2
Output, y'
Input, y
d)
Network 1+2
Output, y'
Input, x
1.0
0.0
-1.0


a)
Output, y
Input, x
b)
Output, y'
Input, y
c)
Output, y'
Input, x

$(D+1)^K$


a)
x_1
x_2
h_1
h_2
h_3
y
h'_1
h'_2
y'
b)
Output, y
Input, x_2
Input, x_1
1.0
0.0
-1.0
c)
Output, y'
Input, y
d)
Output, y'
Input, x_2
Input, x_1



Input, $\mathbf{x}$

$D_i = 3$

Hidden layer, $\mathbf{h}_1$

$D_1 = 4$

Hidden layer, $\mathbf{h}_2$

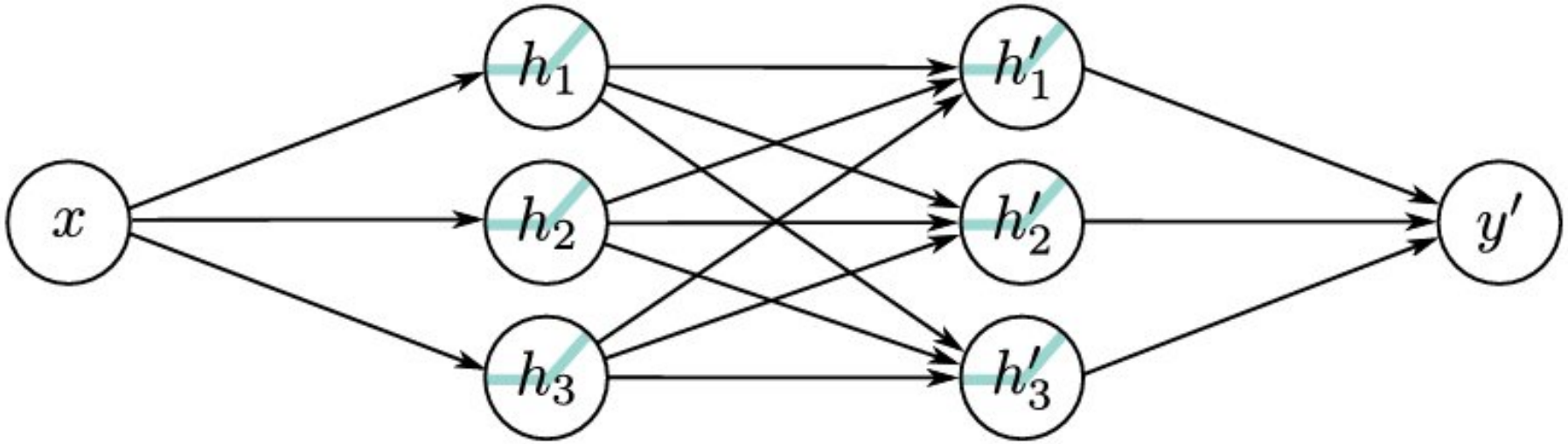
$D_2 = 2$

Hidden layer, $\mathbf{h}_3$

$D_3 = 3$

Output, $\mathbf{y}$

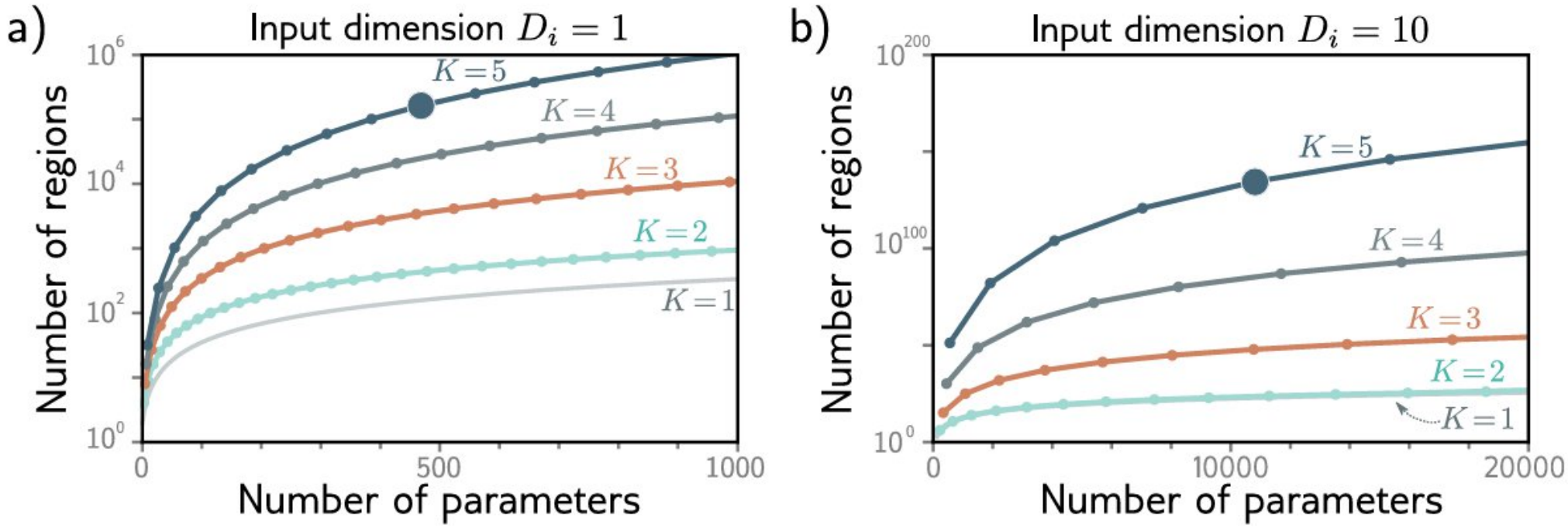
$D_o = 2$



$$
\begin{aligned}
y' = \; & \phi'_0 + \phi'_1 \mathrm{a}\left[\psi_{10} + \psi_{11}\mathrm{a}[\theta_{10} + \theta_{11}x] + \psi_{12}\mathrm{a}[\theta_{20} + \theta_{21}x] + \psi_{13}\mathrm{a}[\theta_{30} + \theta_{31}x]\right] \\
& + \phi'_2 \mathrm{a}[\psi_{20} + \psi_{21}\mathrm{a}[\theta_{10} + \theta_{11}x] + \psi_{22}\mathrm{a}[\theta_{20} + \theta_{21}x] + \psi_{23}\mathrm{a}[\theta_{30} + \theta_{31}x]] \\
& + \phi'_3 \mathrm{a}[\psi_{30} + \psi_{31}\mathrm{a}[\theta_{10} + \theta_{11}x] + \psi_{32}\mathrm{a}[\theta_{20} + \theta_{21}x] + \psi_{33}\mathrm{a}[\theta_{30} + \theta_{31}x]],
\end{aligned}
$$

$$
3D + 1 + (K-1)D(D+1)
$$

D & K AS HYPERPARAMETERS



CHAPTER 5:
LOSS FUNCTIONS

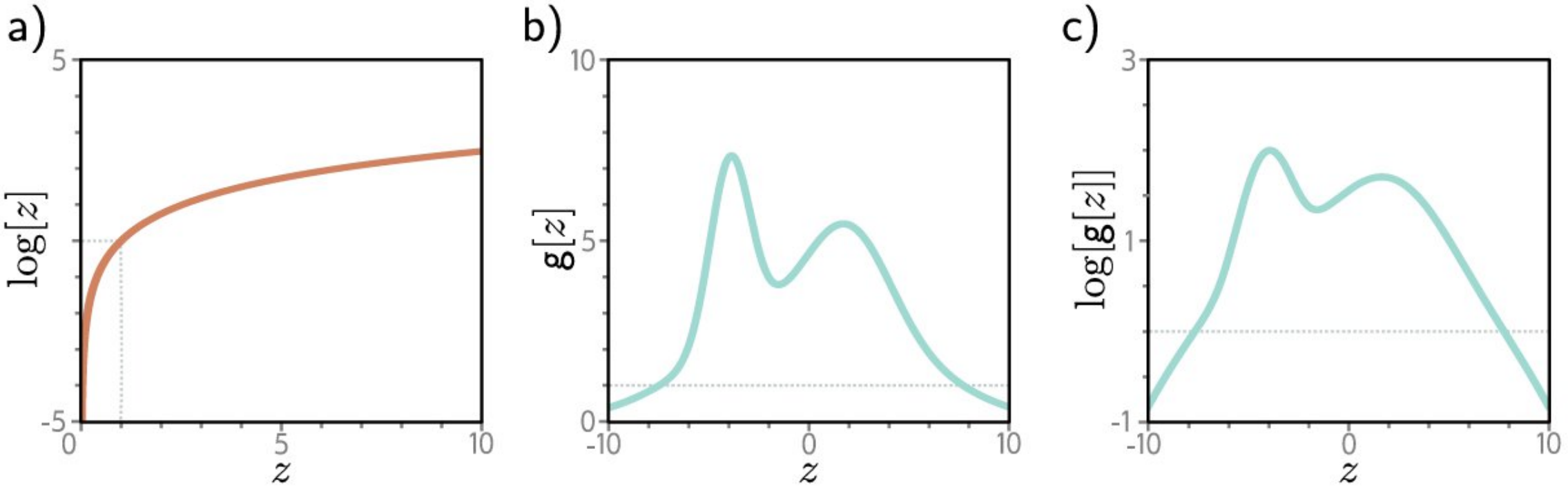
MAXIMUM LIKELIHOOD

$$D = \{(x_1, y_1), \ldots, (x_N, y_N)\}$$

$$L(\theta) = \prod_{i=1}^{N} p(y_i \mid x_i; \theta)$$

$$\theta^* = \arg\max_{\theta} L(\theta)$$

$$\theta^* = \arg\min_{\theta} \left[-\sum_{i=1}^{N} \log p(y_i \mid x_i; \theta) \right]$$



BINARY CROSS ENTROPY

$$\hat{p}_i = P(y_i = 1 \mid x_i)$$

$$L(\theta) = \prod_i \hat{p}_i^{y_i} (1 - \hat{p}_i)^{1 - y_i}$$

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_i [y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i)]$$

MULTICLASS CROSS ENTROPY

$$Pr(y=k|\mathbf{x}) = \text{softmax}_k\Big[\mathbf{f}[\mathbf{x},\boldsymbol{\phi}]\Big] \qquad \text{softmax}_k[\mathbf{z}] = \frac{\exp[z_k]}{\sum_{k'=1}^{K}\exp[z_{k'}]}$$

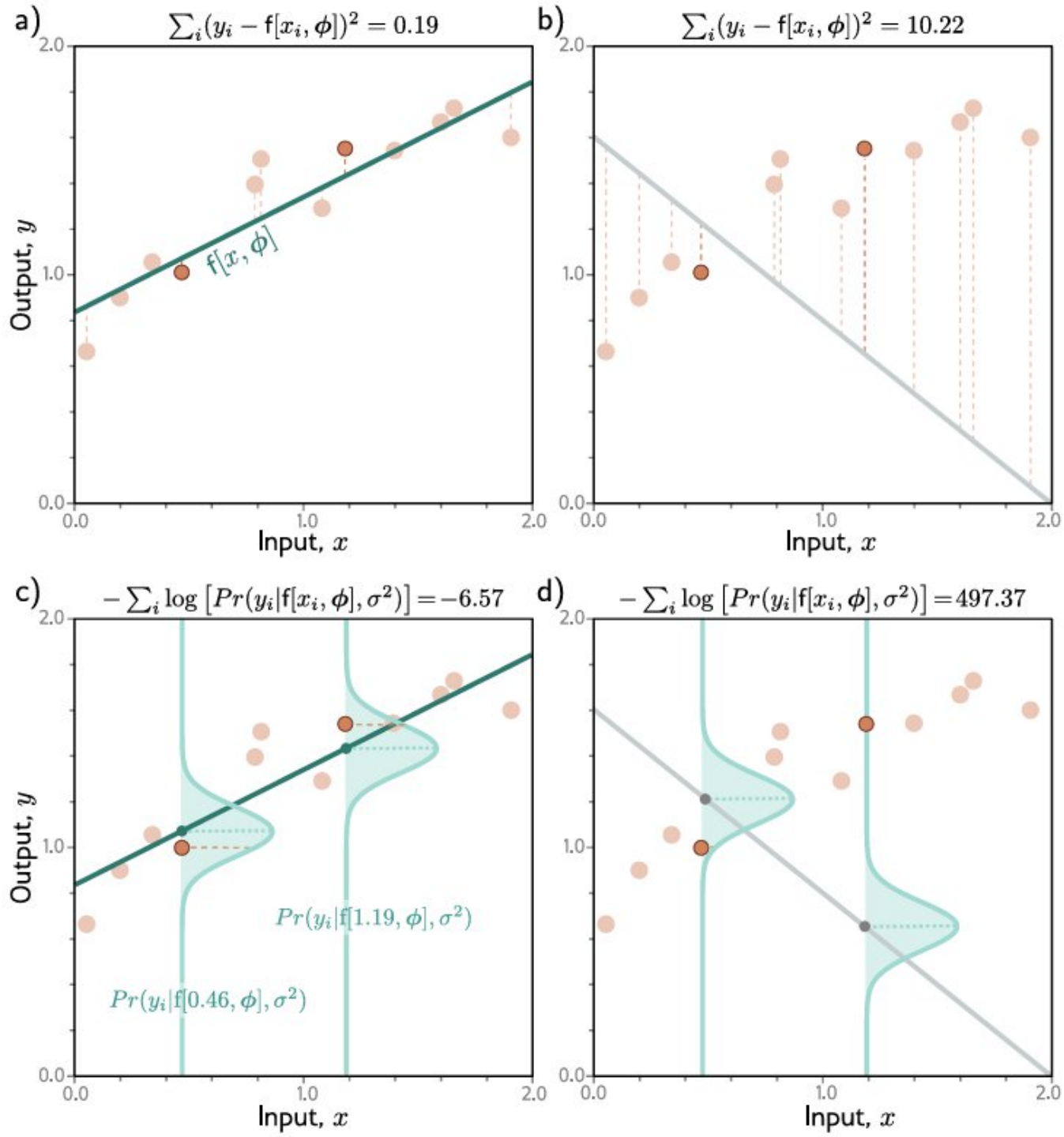
$$\begin{aligned} L[\boldsymbol{\phi}] &= -\sum_{i=1}^{I}\log\Big[\text{softmax}_{y_i}\Big[\mathbf{f}[\mathbf{x}_i,\boldsymbol{\phi}]\Big]\Big] \\ &= -\sum_{i=1}^{I}\left(\text{f}_{y_i}[\mathbf{x}_i,\boldsymbol{\phi}] - \log\left[\sum_{k'=1}^{K}\exp[\,\text{f}_{k'}[\mathbf{x}_i,\boldsymbol{\phi}]]\right]\right) \end{aligned}$$

MEAN SQUARE ERROR

$$Pr(y|\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left[-\frac{(y-\mu)^2}{2\sigma^2}\right]$$

$$Pr(y|\mathrm{f}[\mathbf{x},\boldsymbol{\phi}],\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left[-\frac{(y-\mathrm{f}[\mathbf{x},\boldsymbol{\phi}])^2}{2\sigma^2}\right]$$

$$\begin{aligned}
\hat{\boldsymbol{\phi}} &= \underset{\boldsymbol{\phi}}{\operatorname{argmin}}\left[-\sum_{i=1}^{I}\log\left[\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left[-\frac{(y_i-\mathrm{f}[\mathbf{x}_i,\boldsymbol{\phi}])^2}{2\sigma^2}\right]\right]\right] \\
&= \underset{\boldsymbol{\phi}}{\operatorname{argmin}}\left[-\sum_{i=1}^{I}\log\left[\frac{1}{\sqrt{2\pi\sigma^2}}\right] - \frac{(y_i-\mathrm{f}[\mathbf{x}_i,\boldsymbol{\phi}])^2}{2\sigma^2}\right] \\
&= \underset{\boldsymbol{\phi}}{\operatorname{argmin}}\left[-\sum_{i=1}^{I} -\frac{(y_i-\mathrm{f}[\mathbf{x}_i,\boldsymbol{\phi}])^2}{2\sigma^2}\right] \\
&= \underset{\boldsymbol{\phi}}{\operatorname{argmin}}\left[\sum_{i=1}^{I}(y_i-\mathrm{f}[\mathbf{x}_i,\boldsymbol{\phi}])^2\right],
\end{aligned}$$


a)
$\sum_i (y_i - \mathrm{f}[x_i, \phi])^2 = 0.19$
b)
$\sum_i (y_i - \mathrm{f}[x_i, \phi])^2 = 10.22$
c)
$-\sum_i \log\left[Pr(y_i|\mathrm{f}[x_i, \phi], \sigma^2)\right] = -6.57$
d)
$-\sum_i \log\left[Pr(y_i|\mathrm{f}[x_i, \phi], \sigma^2)\right] = 497.37$
$\mathrm{f}[x, \phi]$
$Pr(y_i|\mathrm{f}[1.19, \phi], \sigma^2)$
$Pr(y_i|\mathrm{f}[0.46, \phi], \sigma^2)$
Output, y
Input, x
0.0
1.0
2.0

MULTIPLE OUTPUTS

$$Pr(\mathbf{y}|\mathbf{f}[\mathbf{x}_i, \boldsymbol{\phi}]) = \prod_d Pr(y_d|\mathbf{f}_d[\mathbf{x}_i, \boldsymbol{\phi}])$$

$$L[\boldsymbol{\phi}] = -\sum_{i=1}^{I} \log\Big[Pr(\mathbf{y}|\mathbf{f}[\mathbf{x}_i, \boldsymbol{\phi}])\Big] = -\sum_{i=1}^{I}\sum_d \log\Big[Pr(y_{id}|\mathbf{f}_d[\mathbf{x}_i, \boldsymbol{\phi}])\Big]$$

Data Type	Domain	Distribution	Use
univariate, continuous, unbounded	$y \in \mathbb{R}$	univariate normal	regression
univariate, continuous, unbounded	$y \in \mathbb{R}$	Laplace or t-distribution	robust regression
univariate, continuous, unbounded	$y \in \mathbb{R}$	mixture of Gaussians	multimodal regression
univariate, continuous, bounded below	$y \in \mathbb{R}^+$	exponential or gamma	predicting magnitude
univariate, continuous, bounded	$y \in [0, 1]$	beta	predicting proportions
multivariate, continuous, unbounded	$\mathbf{y} \in \mathbb{R}^K$	multivariate normal	multivariate regression
univariate, continuous, circular	$y \in (-\pi, \pi]$	von Mises	predicting direction
univariate, discrete, binary	$y \in \{0, 1\}$	Bernoulli	binary classification
univariate, discrete, bounded	$y \in \{1, 2, \ldots, K\}$	categorical	multiclass classification
univariate, discrete, bounded below	$y \in [0, 1, 2, 3, \ldots]$	Poisson	predicting event counts
multivariate, discrete, permutation	$\mathbf{y} \in \text{Perm}[1, 2, \ldots, K]$	Plackett-Luce	ranking

Figure 5.11 Distributions for loss functions for different prediction types.

CHAPTER 6:
FITTING MODELS

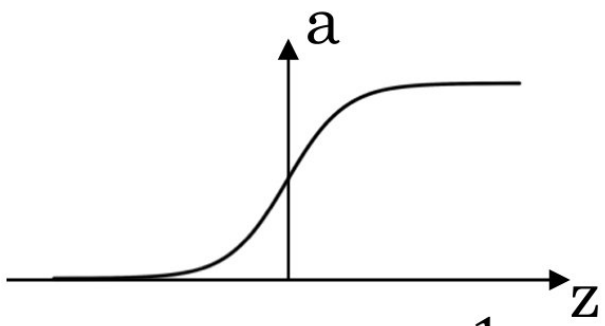
GRADIENT DESCENT

Step 1. Compute the derivatives of the loss with respect to the parameters:

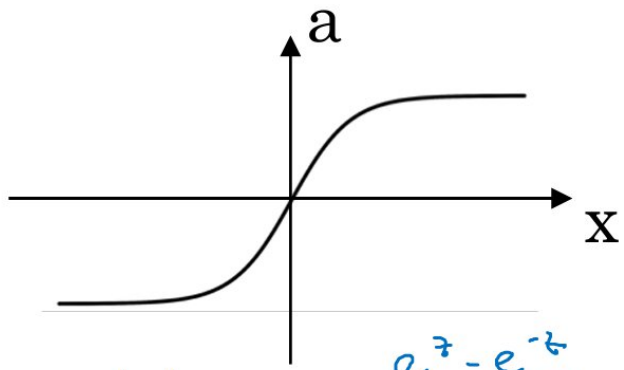
$$\frac{\partial L}{\partial \boldsymbol{\phi}} = \begin{bmatrix} \frac{\partial L}{\partial \phi_0} \\ \frac{\partial L}{\partial \phi_1} \\ \vdots \\ \frac{\partial L}{\partial \phi_N} \end{bmatrix}. \tag{6.2}$$

Step 2. Update the parameters according to the rule:

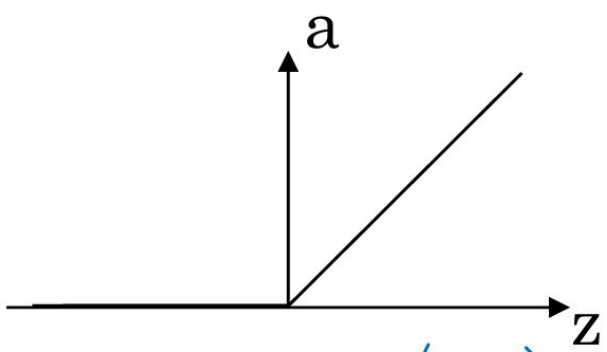
$$\boldsymbol{\phi} \longleftarrow \boldsymbol{\phi} - \alpha \cdot \frac{\partial L}{\partial \boldsymbol{\phi}}, \tag{6.3}$$



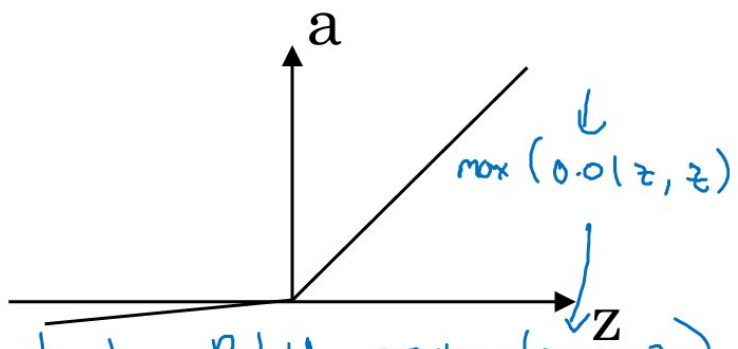
sigmoid: $a = \frac{1}{1+e^{-z}}$



tanh: $a = \frac{e^{z}-e^{-z}}{e^{z}+e^{-z}}$



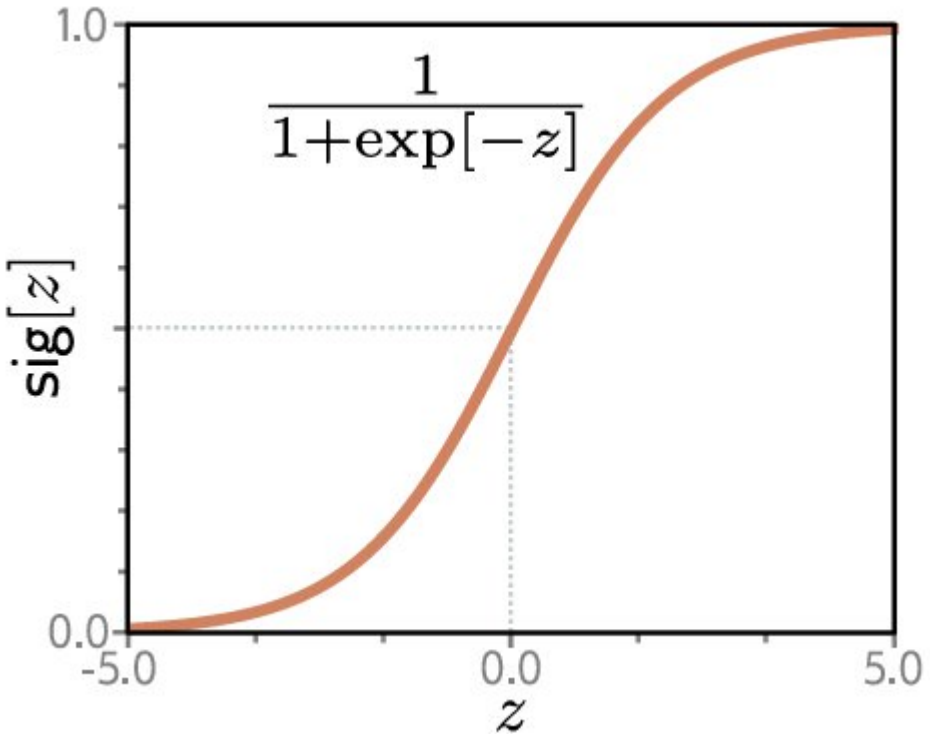
ReLU $a = \max(0, z)$



Leaky ReLU $a = \max(0.01z, z)$

Andrew Ng

SIGMOID FUNCTION

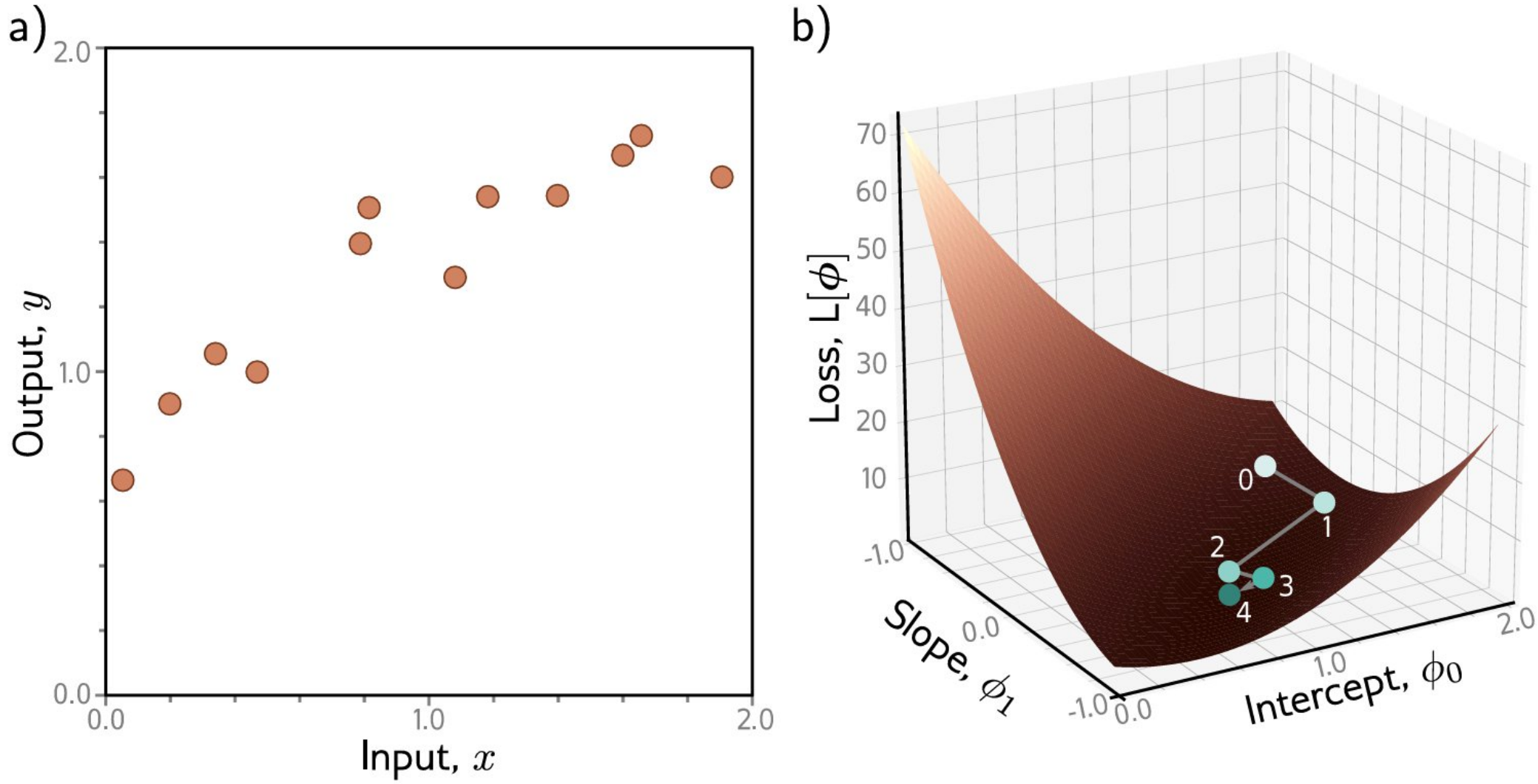


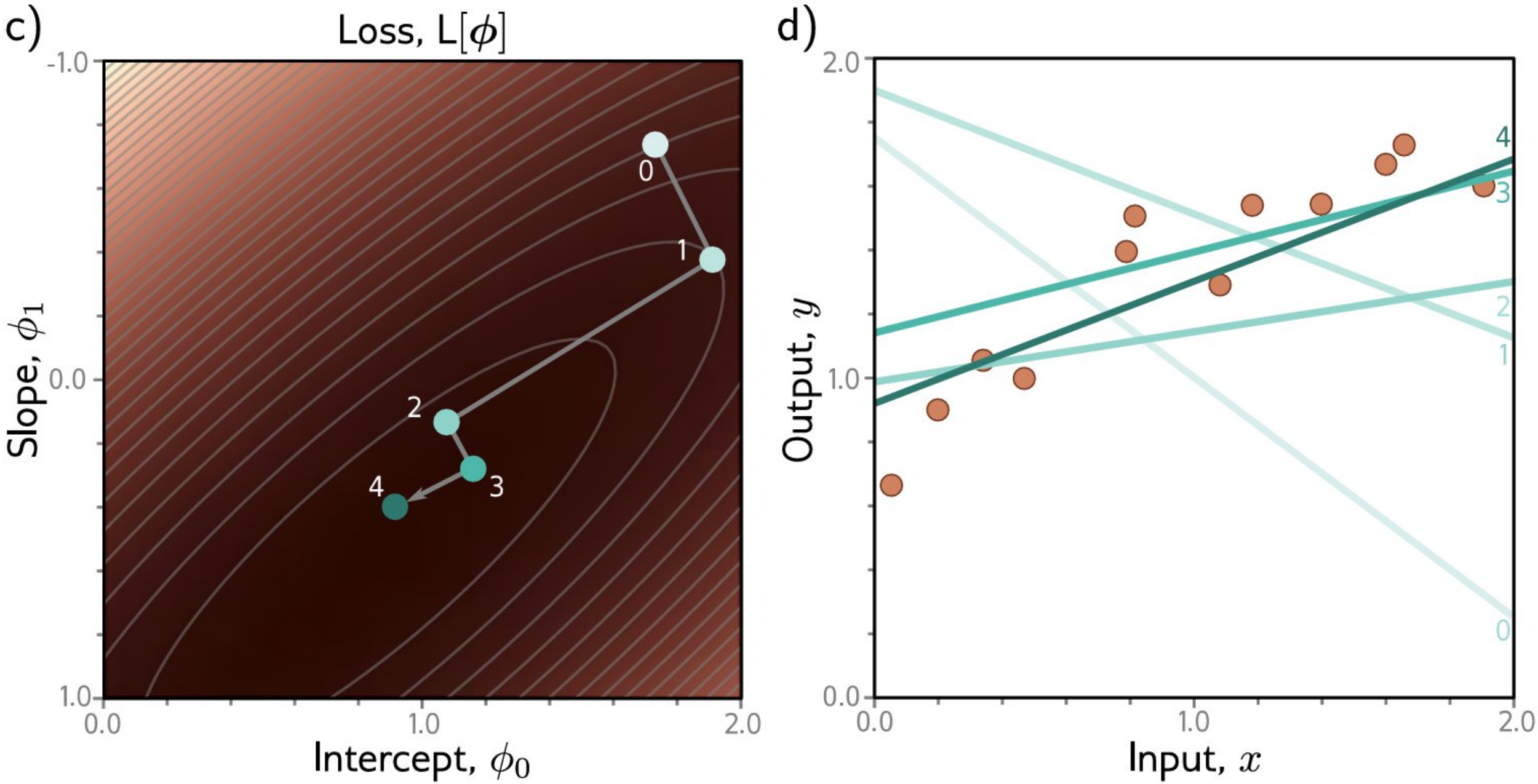
$$L[\boldsymbol{\phi}] = \sum_{i=1}^{I} -(1-y_i)\log\Big[1-\text{sig}[\text{f}[\mathbf{x}_i,\boldsymbol{\phi}]]\Big] - y_i\log\Big[\text{sig}[\text{f}[\mathbf{x}_i,\boldsymbol{\phi}]]\Big]$$

LINEAR REGRESSION EXAMPLE

$$\begin{aligned} \mathrm{L}[\boldsymbol{\phi}] &= \sum_{i=1}^{I} \ell_i = \sum_{i=1}^{I} \left(\mathrm{f}[x_i, \boldsymbol{\phi}] - y_i\right)^2 \\ &= \sum_{i=1}^{I} \left(\phi_0 + \phi_1 x_i - y_i\right)^2, \end{aligned}$$

LINEAR REGRESSION - CONVEX- GLOBAL MINIMUM



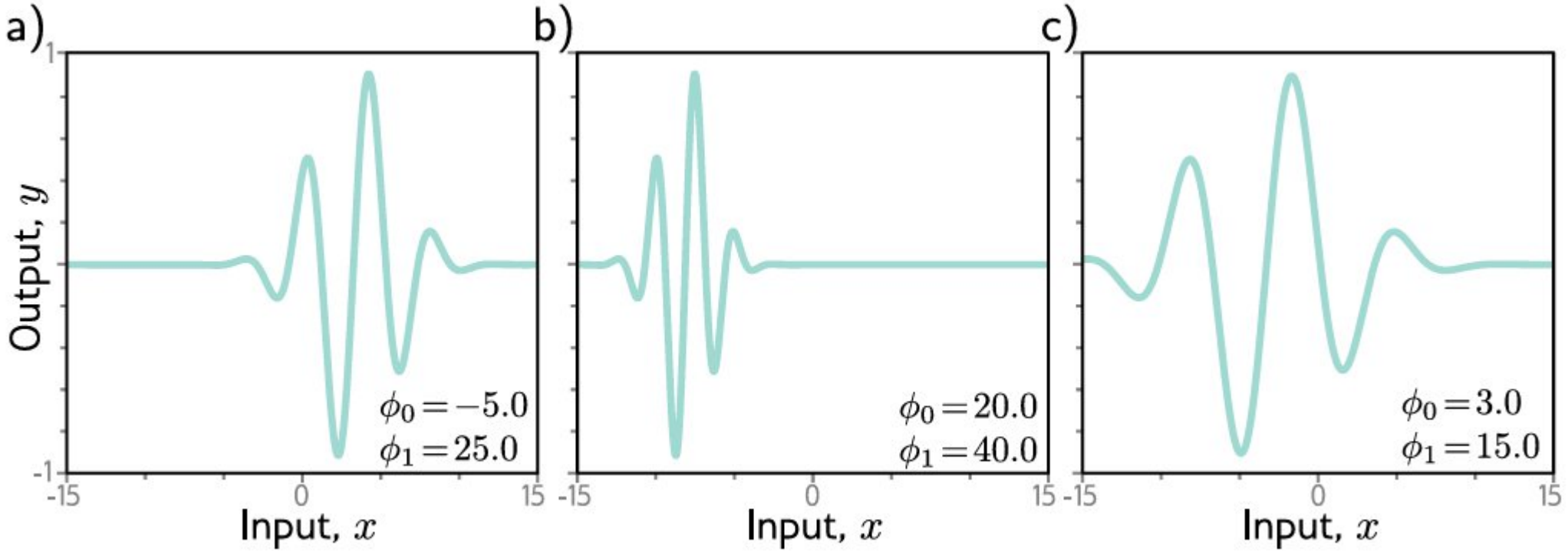

c)
Loss, L[ϕ]
-1.0
0.0
1.0
Slope, ϕ_1
0.0
1.0
2.0
Intercept, ϕ_0
0
1
2
3
4
d)
2.0
1.0
0.0
Output, y
0.0
1.0
2.0
Input, x
4
3
2
1
0

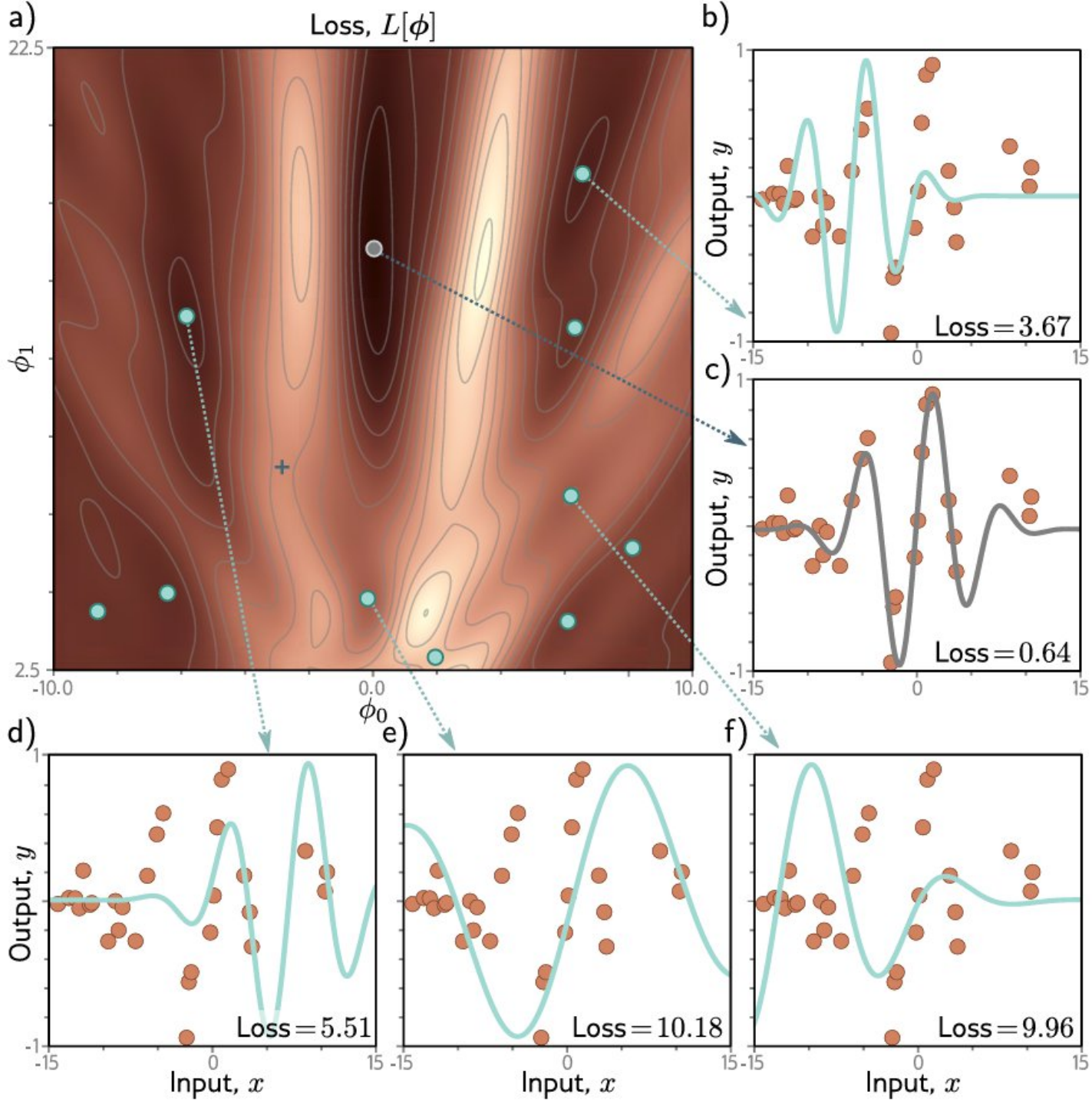
$$\frac{\partial L}{\partial \boldsymbol{\phi}} = \frac{\partial}{\partial \boldsymbol{\phi}} \sum_{i=1}^{I} \ell_i = \sum_{i=1}^{I} \frac{\partial \ell_i}{\partial \boldsymbol{\phi}},$$

$$\frac{\partial \ell_i}{\partial \boldsymbol{\phi}} = \begin{bmatrix} \frac{\partial \ell_i}{\partial \phi_0} \\ \frac{\partial \ell_i}{\partial \phi_1} \end{bmatrix} = \begin{bmatrix} 2(\phi_0 + \phi_1 x_i - y_i) \\ 2x_i(\phi_0 + \phi_1 x_i - y_i) \end{bmatrix}$$

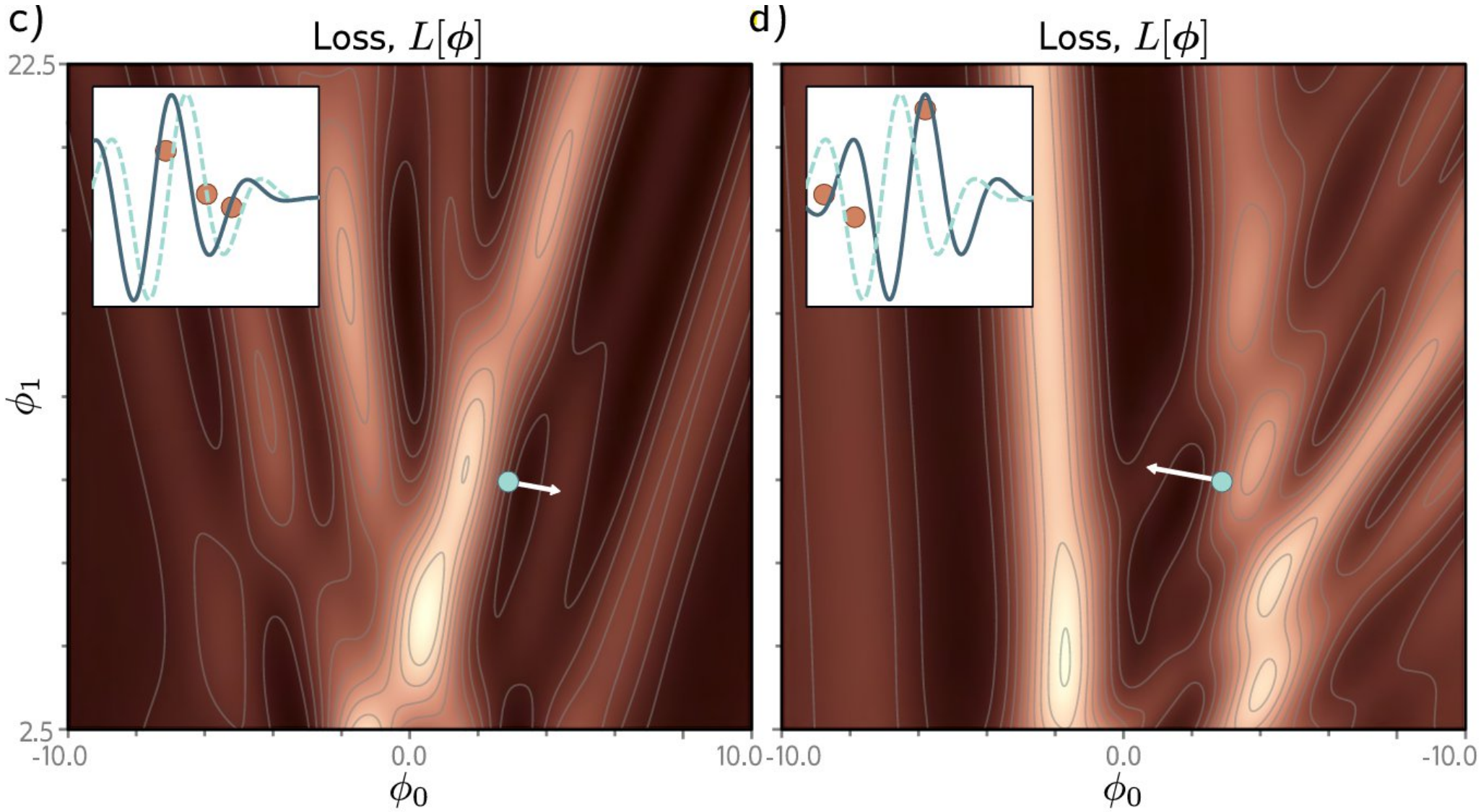
NON-LINEAR EXAMPLE

$$\mathrm{f}[x, \boldsymbol{\phi}] = \sin[\phi_0 + 0.06 \cdot \phi_1 x] \cdot \exp\left(-\frac{(\phi_0 + 0.06 \cdot \phi_1 x)^2}{32.0}\right)$$




a)
Loss, $L[\phi]$
22.5
2.5
ϕ_1
-10.0
0.0
10.0
ϕ_0
b)
Output, y
Loss = 3.67
c)
Output, y
Loss = 0.64
d)
Output, y
Loss = 5.51
Input, x
e)
Loss = 10.18
Input, x
f)
Loss = 9.96
Input, x

STOCHASTIC GRADIENT DESCENT



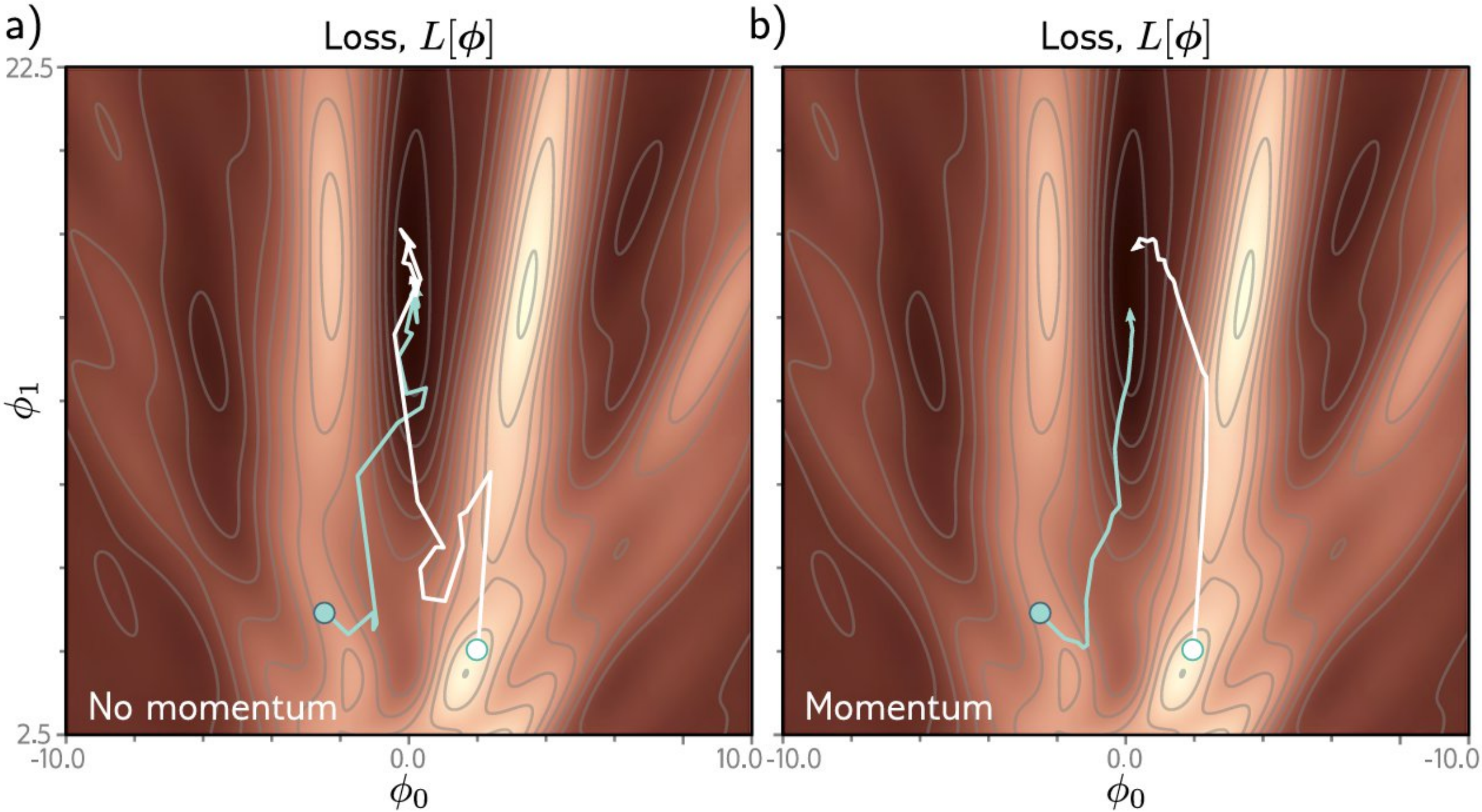
SGD’S PROS

- Computation cost
- Chance of escaping saddle points
- Chance of escaping local minimum

MOMENTUM

$$\mathbf{m}_{t+1} \leftarrow \beta \cdot \mathbf{m}_t + (1-\beta) \sum_{i \in \mathcal{B}_t} \frac{\partial \ell_i[\boldsymbol{\phi}_t]}{\partial \boldsymbol{\phi}}$$

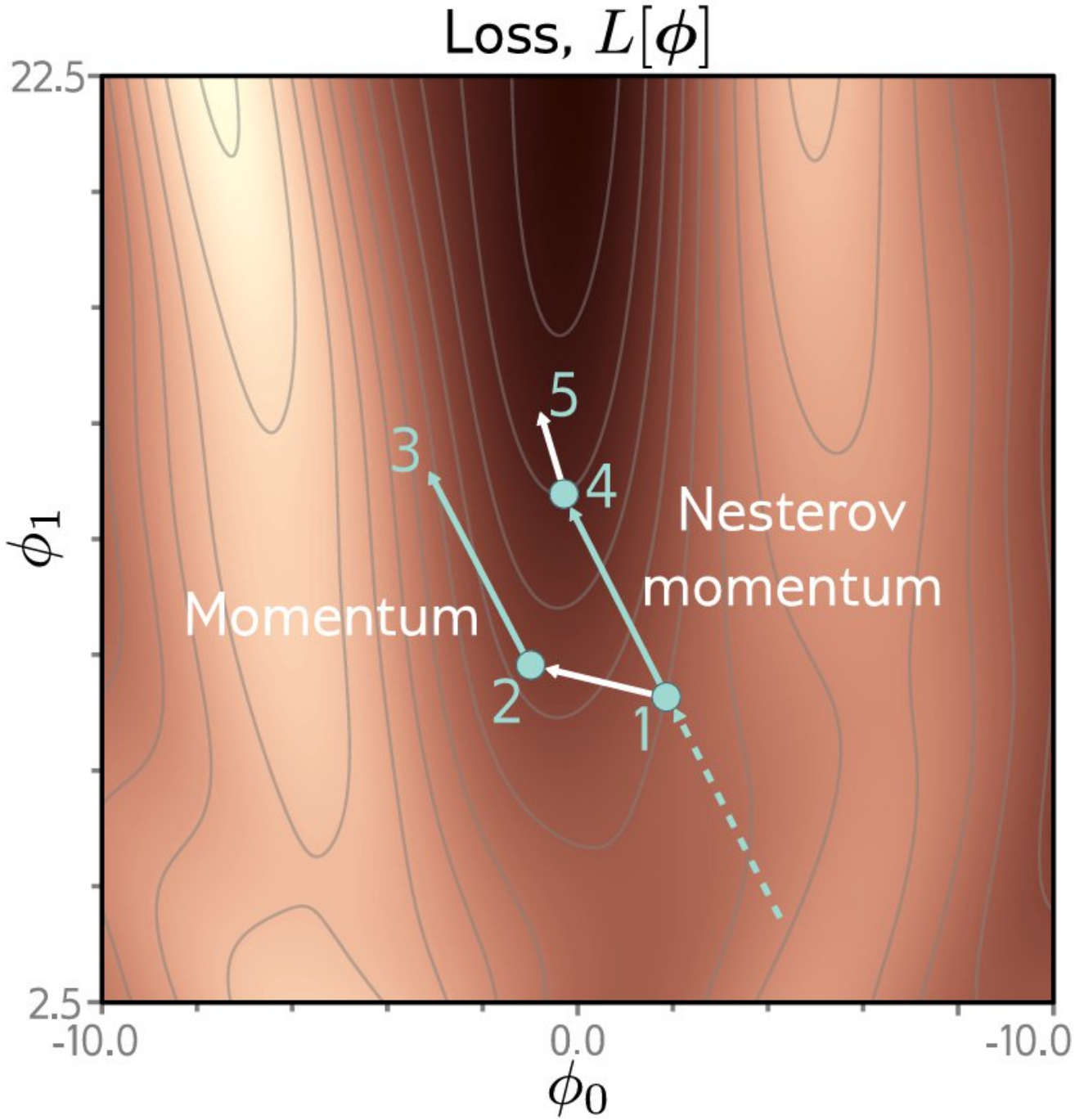
$$\boldsymbol{\phi}_{t+1} \leftarrow \boldsymbol{\phi}_t - \alpha \cdot \mathbf{m}_{t+1},$$


a)
Loss, $L[\phi]$
22.5
2.5
ϕ_1
No momentum
-10.0
0.0
10.0
ϕ_0
b)
Loss, $L[\phi]$
Momentum
-10.0
0.0
-10.0
ϕ_0

NESTEROV ACCELERATED MOMENTUM

$$\mathbf{m}_{t+1} \leftarrow \beta \cdot \mathbf{m}_t + (1-\beta) \sum_{i \in \mathcal{B}_t} \frac{\partial \ell_i [\boldsymbol{\phi}_t - \alpha \cdot \mathbf{m}_t]}{\partial \boldsymbol{\phi}}$$

$$\boldsymbol{\phi}_{t+1} \leftarrow \boldsymbol{\phi}_t - \alpha \cdot \mathbf{m}_{t+1},$$


Loss, $L[\phi]$
22.5
2.5
ϕ_1
-10.0
0.0
-10.0
ϕ_0
1
2
3
4
5
Momentum
Nesterov
momentum

NORMALIZED GRADIENT DESCENT

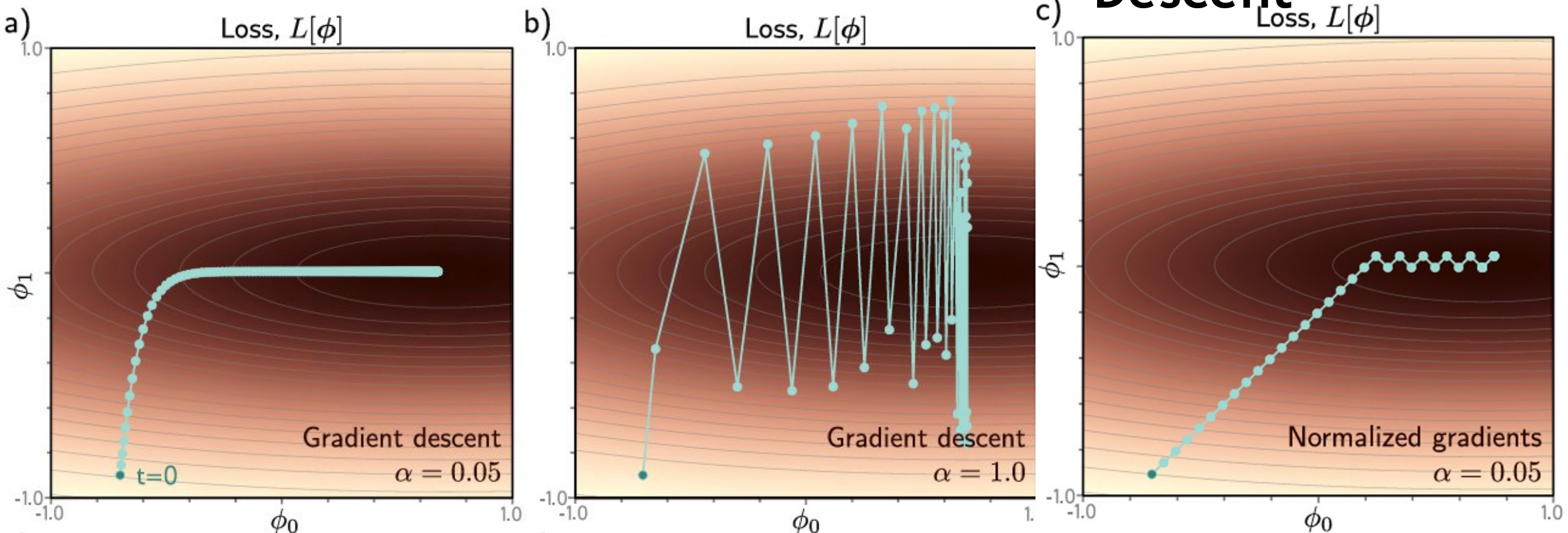
$$\mathbf{m}_{t+1} \leftarrow \frac{\partial L[\boldsymbol{\phi}_t]}{\partial \boldsymbol{\phi}}$$

$$\mathbf{v}_{t+1} \leftarrow \left(\frac{\partial L[\boldsymbol{\phi}_t]}{\partial \boldsymbol{\phi}}\right)^2$$

$$\boldsymbol{\phi}_{t+1} \leftarrow \boldsymbol{\phi}_t - \alpha \cdot \frac{\mathbf{m}_{t+1}}{\sqrt{\mathbf{v}_{t+1}} + \epsilon}$$

Gradient Descent

Normalized Gradient Descent



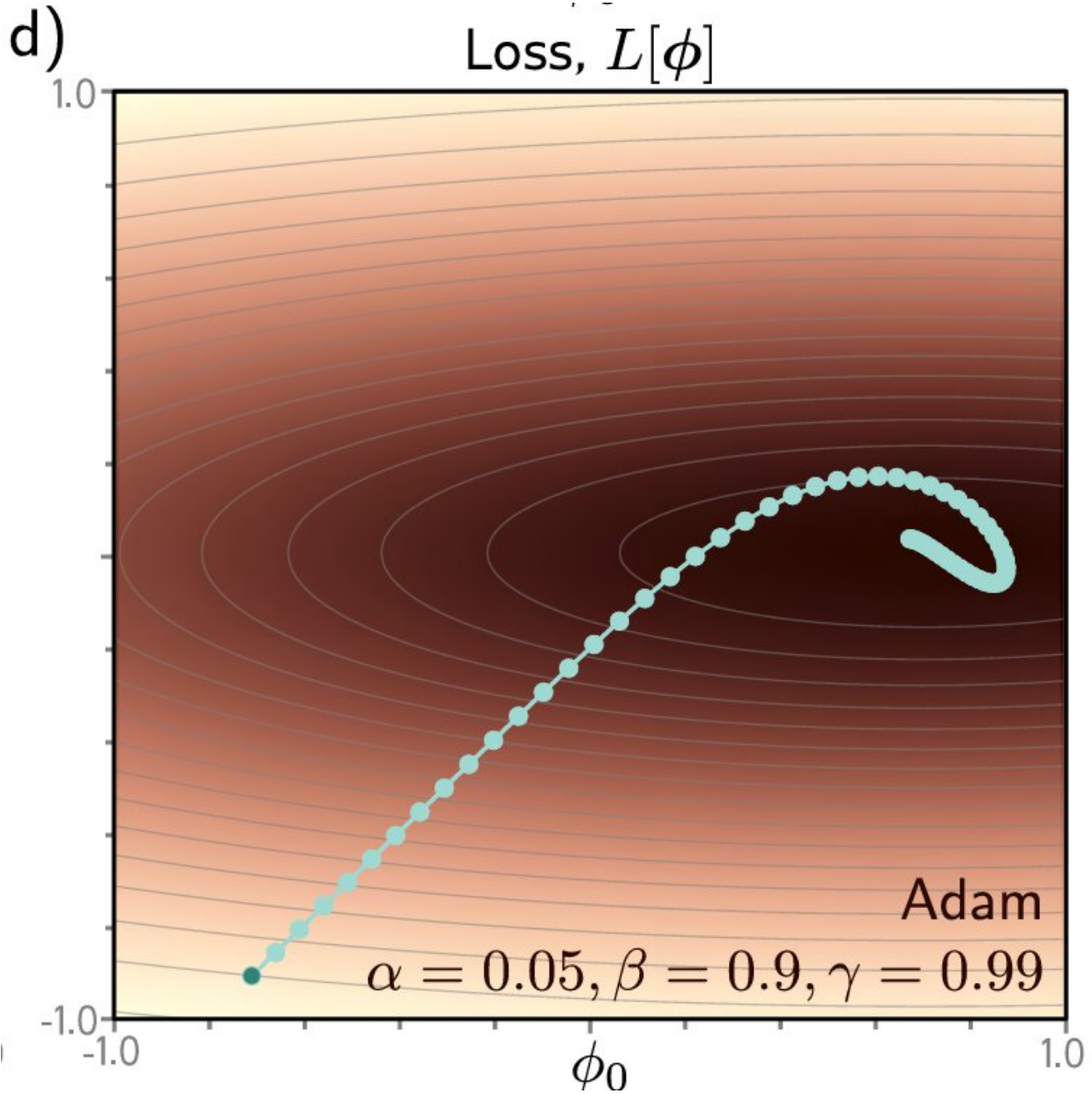
ADAPTIVE MOMENT ESTIMATION

$$\mathbf{m}_{t+1} \leftarrow \beta \cdot \mathbf{m}_t + (1-\beta)\frac{\partial L[\boldsymbol{\phi}_t]}{\partial \boldsymbol{\phi}}$$

$$\mathbf{v}_{t+1} \leftarrow \gamma \cdot \mathbf{v}_t + (1-\gamma)\left(\frac{\partial L[\boldsymbol{\phi}_t]}{\partial \boldsymbol{\phi}}\right)^2$$

$$\tilde{\mathbf{m}}_{t+1} \leftarrow \frac{\mathbf{m}_{t+1}}{1-\beta^{t+1}}$$

$$\tilde{\mathbf{v}}_{t+1} \leftarrow \frac{\mathbf{v}_{t+1}}{1-\gamma^{t+1}}$$



$$\phi_{t+1} \leftarrow \phi_t - \alpha \cdot \frac{\tilde{\mathbf{m}}_{t+1}}{\sqrt{\tilde{\mathbf{v}}_{t+1}} + \epsilon}$$

ADAM IS USUALLY COMPUTED WITH MINI-BACH

$$\mathbf{m}_{t+1} \leftarrow \beta \cdot \mathbf{m}_t + (1-\beta) \sum_{i \in \mathcal{B}_t} \frac{\partial \ell_i[\boldsymbol{\phi}_t]}{\partial \boldsymbol{\phi}}$$

$$\mathbf{v}_{t+1} \leftarrow \gamma \cdot \mathbf{v}_t + (1-\gamma) \sum_{i \in \mathcal{B}_t} \left(\frac{\partial \ell_i[\boldsymbol{\phi}_t]}{\partial \boldsymbol{\phi}} \right)^2$$

HESSIAN MATRIX – CONVEXITY – POSITIVE EIGENVALUES

$$\mathbf{H}[\phi] = \begin{bmatrix} \frac{\partial^2 L}{\partial \phi_0^2} & \frac{\partial^2 L}{\partial \phi_0 \partial \phi_1} & \cdots & \frac{\partial^2 L}{\partial \phi_0 \partial \phi_N} \\ \frac{\partial^2 L}{\partial \phi_1 \partial \phi_0} & \frac{\partial^2 L}{\partial \phi_1^2} & \cdots & \frac{\partial^2 L}{\partial \phi_1 \partial \phi_N} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 L}{\partial \phi_N \partial \phi_0} & \frac{\partial^2 L}{\partial \phi_N \partial \phi_1} & \cdots & \frac{\partial^2 L}{\partial \phi_N^2} \end{bmatrix}$$

CHAPTER 7:
GRADIENTS & INITIALIZATION

BACKPROPAGATION

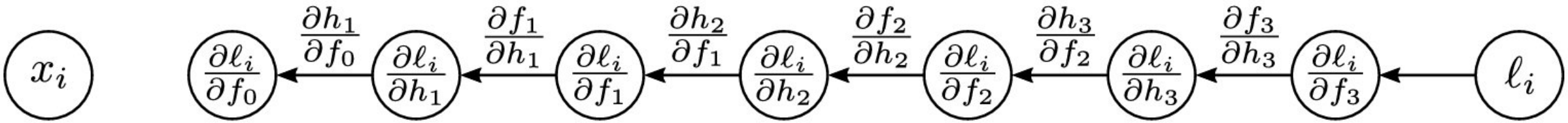
$$
\begin{aligned}
f_0 &= \beta_0 + \omega_0 \cdot x_i \\
h_1 &= \sin[f_0] \\
f_1 &= \beta_1 + \omega_1 \cdot h_1 \\
h_2 &= \exp[f_1] \\
f_2 &= \beta_2 + \omega_2 \cdot h_2 \\
h_3 &= \cos[f_2] \\
f_3 &= \beta_3 + \omega_3 \cdot h_3 \\
\ell_i &= (f_3 - y_i)^2.
\end{aligned}
$$

$$\mathrm{L}[\boldsymbol{\phi}] = \sum_i \ell_i$$

$$\ell_i = \left(\mathrm{f}[x_i, \boldsymbol{\phi}] - y_i\right)^2$$

$$\frac{\partial \ell_i}{\partial \beta_0}, \quad \frac{\partial \ell_i}{\partial \omega_0}, \quad \frac{\partial \ell_i}{\partial \beta_1}, \quad \frac{\partial \ell_i}{\partial \omega_1}, \quad \frac{\partial \ell_i}{\partial \beta_2}, \quad \frac{\partial \ell_i}{\partial \omega_2}, \quad \frac{\partial \ell_i}{\partial \beta_3}, \quad \text{and} \quad \frac{\partial \ell_i}{\partial \omega_3}$$

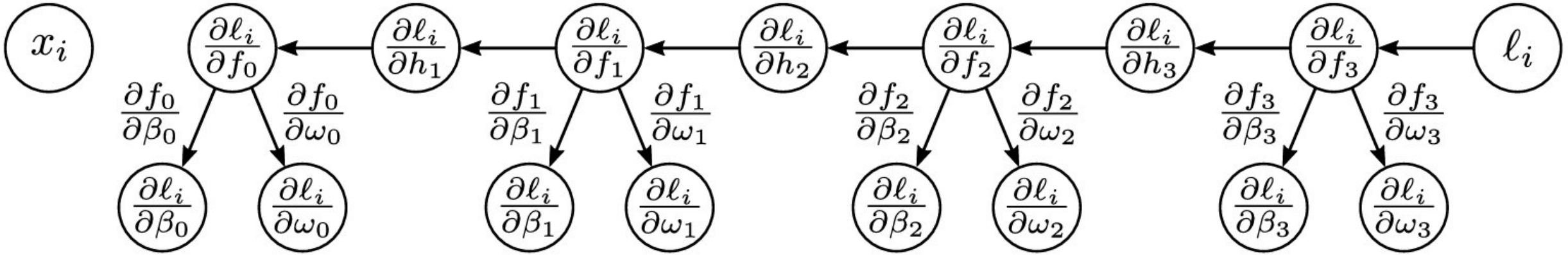
FORWARD PASS



$$
\begin{aligned}
\frac{\partial \ell_i}{\partial f_2} &= \frac{\partial h_3}{\partial f_2}\left(\frac{\partial f_3}{\partial h_3}\frac{\partial \ell_i}{\partial f_3}\right) \\
\frac{\partial \ell_i}{\partial h_2} &= \frac{\partial f_2}{\partial h_2}\left(\frac{\partial h_3}{\partial f_2}\frac{\partial f_3}{\partial h_3}\frac{\partial \ell_i}{\partial f_3}\right) \\
\frac{\partial \ell_i}{\partial f_1} &= \frac{\partial h_2}{\partial f_1}\left(\frac{\partial f_2}{\partial h_2}\frac{\partial h_3}{\partial f_2}\frac{\partial f_3}{\partial h_3}\frac{\partial \ell_i}{\partial f_3}\right) \\
\frac{\partial \ell_i}{\partial h_1} &= \frac{\partial f_1}{\partial h_1}\left(\frac{\partial h_2}{\partial f_1}\frac{\partial f_2}{\partial h_2}\frac{\partial h_3}{\partial f_2}\frac{\partial f_3}{\partial h_3}\frac{\partial \ell_i}{\partial f_3}\right) \\
\frac{\partial \ell_i}{\partial f_0} &= \frac{\partial h_1}{\partial f_0}\left(\frac{\partial f_1}{\partial h_1}\frac{\partial h_2}{\partial f_1}\frac{\partial f_2}{\partial h_2}\frac{\partial h_3}{\partial f_2}\frac{\partial f_3}{\partial h_3}\frac{\partial \ell_i}{\partial f_3}\right)
\end{aligned}
$$

$$
\begin{aligned}
f_0 &= \beta_0 + \omega_0 \cdot x_i \\
h_1 &= \sin[f_0] \\
f_1 &= \beta_1 + \omega_1 \cdot h_1 \\
h_2 &= \exp[f_1] \\
f_2 &= \beta_2 + \omega_2 \cdot h_2 \\
h_3 &= \cos[f_2] \\
f_3 &= \beta_3 + \omega_3 \cdot h_3 \\
\ell_i &= (f_3 - y_i)^2.
\end{aligned}
$$

BACKWARD PASS



$$\frac{\partial \ell_i}{\partial \beta_k} = \frac{\partial f_k}{\partial \beta_k}\frac{\partial \ell_i}{\partial f_k}$$

$$\frac{\partial \ell_i}{\partial \omega_k} = \frac{\partial f_k}{\partial \omega_k}\frac{\partial \ell_i}{\partial f_k}$$

$$\mathbf{f}_1 = \boldsymbol{\beta}_1 + \boldsymbol{\Omega}_1\mathbf{h}_1$$